

Task 1: Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

```
shreyas@Grigori:~$ mkdir -p ~/producer_consumer_pipe
shreyas@Grigori:~$ cd ~/producer_consumer_pipe
shreyas@Grigori:~/producer_consumer_pipe$ nano pipe.c
shreyas@Grigori:~/producer_consumer_pipe$ gcc pipe.c -o pipe
shreyas@Grigori:~/producer_consumer_pipe$ ls
pipe  pipe.c
shreyas@Grigori:~/producer_consumer_pipe$ ./pipe
Child Consumer:
Messages consumed : 100000
Total bytes      : 3888895

=====
Producer-Consumer Pipe Communication
=====
Messages transferred : 100000
Time taken           : 0.130033 seconds
Communication rate    : 769035.55 messages/sec
=====
```

Task 2: Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
shreyas@Grigori: ~/ls_grep_p × + v
shreyas@Grigori:~$ mkdir -p ~/ls_grep_pipe
shreyas@Grigori:~$ cd ~/ls_grep_pipe
shreyas@Grigori:~/ls_grep_pipe$ nano ls_grep.c
shreyas@Grigori:~/ls_grep_pipe$ gcc ls_grep.c -o ls_grep
shreyas@Grigori:~/ls_grep_pipe$ ls
ls_grep  ls_grep.c
shreyas@Grigori:~/ls_grep_pipe$ ./ls_grep
-rw-r--r-- 1 shreyas shreyas 1499 Sep 20 14:32 ls_grep.c
shreyas@Grigori:~/ls_grep_pipe$ touch test1.c test2.c hello.c
shreyas@Grigori:~/ls_grep_pipe$ ./ls_grep
-rw-r--r-- 1 shreyas shreyas 0 Sep 20 14:43 hello.c
-rw-r--r-- 1 shreyas shreyas 1499 Sep 20 14:32 ls_grep.c
-rw-r--r-- 1 shreyas shreyas 0 Sep 20 14:43 test1.c
-rw-r--r-- 1 shreyas shreyas 0 Sep 20 14:43 test2.c
shreyas@Grigori:~/ls_grep_pipe$ ls -l | grep ".c"
-rw-r--r-- 1 shreyas shreyas 0 Sep 20 14:43 hello.c
-rw-r--r-- 1 shreyas shreyas 1499 Sep 20 14:32 ls_grep.c
-rw-r--r-- 1 shreyas shreyas 0 Sep 20 14:43 test1.c
-rw-r--r-- 1 shreyas shreyas 0 Sep 20 14:43 test2.c
shreyas@Grigori:~/ls_grep_pipe$ |
```